

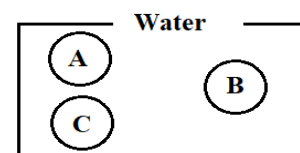
Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ A/E	First Test - Modeule D	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **8 Marks**

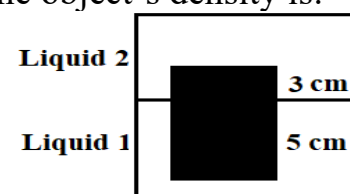
1) Sinking object has upward acceleration		
2) Electric flux depends on the charge inside		
3) Partially and fully immersed is related to Archimedes Principle		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) More electric field lines directed inward				
a) - 1 C	b) 4 C	c) - 7 C	d) 9 C	e) None of these
2) The fully immersed object with a ratio of fluid to object densities 0.8 is:				
a) Floating	b) Sinking	c) At rest	d) Moving up	e) None of these
3) 13 protons and 7 neutrons. The total charge is:				
a) 13 e	b) 7 e	c) - 13 e	d) - 7 e	e) None of these
4) From the figure shown below. The pressures				



a) $P_C > P_A$	b) $P_C < P_B$	c) $P_B = P_C$	d) $P_B > P_C$	e) None of these
5) A cube of x each side is placed in $\vec{E} = \alpha \hat{i} + \beta \hat{j} - \gamma \hat{k}$. The electric flux in xz and xy planes is:				
a) $(\alpha + \beta) x$	b) $(\alpha - \gamma) x^2$	c) $(\beta - \alpha) x^2$	d) $(\beta - \gamma) x^2$	e) None of these
6) An object of ρ suspended between liquid 1 of ρ_1 and liquid 2 of ρ_2 . The object's density is:				



a) $\rho = \frac{3}{8} \rho_1 + \frac{5}{8} \rho_2$	b) $\rho = \frac{5}{8} \rho_1 + \frac{3}{8} \rho_2$	c) $\rho = 5 \rho_1 + 3 \rho_2$	d) $\rho = 3 \rho_1 + 5 \rho_2$	e) None of these
7) The electric flux is given by:				
a) E/A	b) q_{in}/A	c) $EA \cos\theta$	d) $q_{in}/\epsilon_0 \cos\theta$	e) None of these

Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

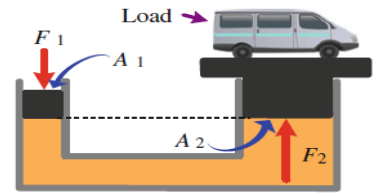
20 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31} \text{ kg}$	$K_e = 9 \times 10^9 \text{ N.m}^2/\text{C}^2$	Gravity: $g = 10 \text{ m/s}^2$
Proton: $m = 1.67 \times 10^{-27} \text{ kg}$	$G = 6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$	Water: $\rho_f = 1000 \text{ kg/m}^3$

1) 500 N object floats in water of $\rho_f = 1000 \text{ kg/m}^3$, the object is immersed by 5 cm and 11 cm over the water. The density of the object in kg/m^3 is:

a) 0.16	b) 454.5	c) 0.117	d) 312.5	e) None of these
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2) In Pascal principle: The piston areas are 25 cm^2 and 750 cm^2 respectively, the weight on the input piston is 1000 N. The weight that the output piston handle is:

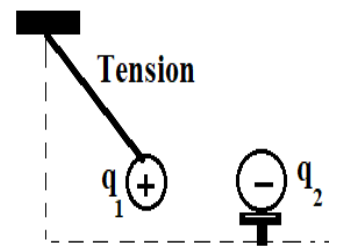


a) 33.3 N	b) 30000 N	c) 333.3 N	d) 300000 N	e) None of these
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3) $q_1 = 2 \text{ nC}$ is placed at $(-0.3 \text{ m}, 0)$, $q_2 = 2 \text{ nC}$ is placed at $(0, 0.3 \text{ m})$. The electric field in N/C at the origin is:

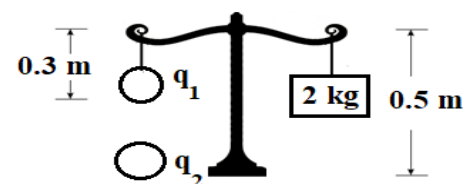
a) 282.8 at 315°	b) 424.3 at 135°	c) 424.3 at 225°	d) 282.8 at 45°	e) None of these
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4) 100 g and $q_1 = 2 \text{ } \mu\text{C}$ charge hung by a rope. A charge of $q_2 = -9 \text{ } \mu\text{C}$ attracts q_1 charge. Then the rope inclined and the distance between the charges is 20 cm. The tension in the rope is:



a) 3.7 N	b) 4.2 N	c) 2.9 N	d) 120.6	e) None of these
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5) 2 kg at one side of a balance scale and a charge $q_1 = 6 \text{ nC}$ at the other as shown. The value of the charge q_2 for equilibrium is:



a) - 0.0444 C	b) 0.01778 C	c) - 0.01481 C	d) 0.0296 C	e) None of these
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